



信号和系统1、2班

群号: 820076205



Signals and Systems

Southern University of Science and Technology

Autumn 2024

陈霏

副教授



陈霏博士于南京大学获得学士和硕士学位，并于2005年于香港中文大学电子工程系毕业获博士学位，其后在美国德州大学达拉斯分校进行博士后研究工作，参与多项NIH资助的大型研究项目，并获美国全国听力研究协会2011年度杰出研究奖。陈霏博士2012-2014年期间在香港大学言语与听觉科学部担任研究助理教授，在<美国声学学报>等国际权威期刊发表了30余篇期刊文章，并曾担任多个IEEE生物医学工程国际学术会议的组委会成员。

研究领域：

- ◆ 言语识别和助听技术
- ◆ 人机接口技术
- ◆ 生物医学信号处理

工作经历：

- ◆ 2014年12月- 至今 南方科技大学 电子与电气工程系 副教授

香港大学 言语和听力科学部 荣誉助理教授

- ◆ 2012年01月- 2014年11月 香港大学 言语和听力科学部 助理教授[研究]

信息工程专业 Information Engineering

- Room: 工学院南座429
- Email: fchen@sustech.edu.cn
- Office hour: Tuesday
10:00-12:00

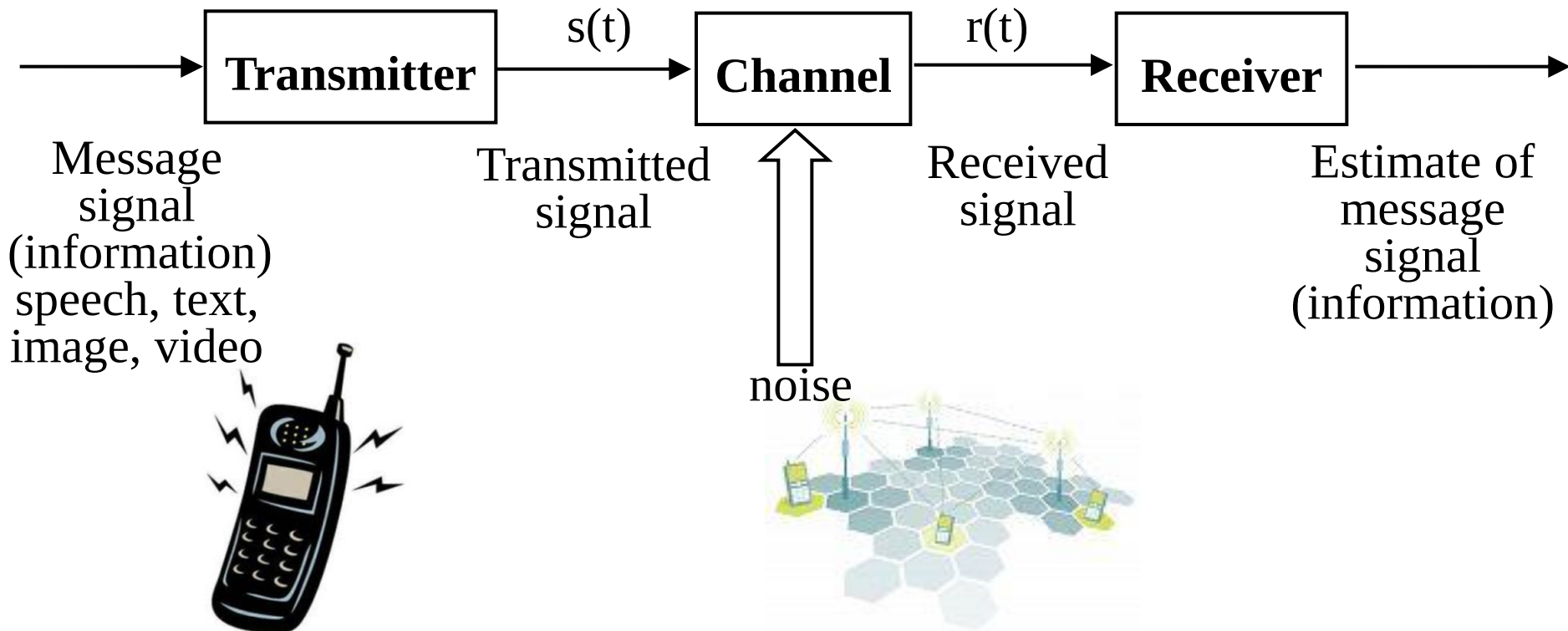
Signals and Systems

- The course is about using **mathematical** techniques to help analyze and synthesize systems which process signals.
 - ◆ Signals are variables that carry information.
 - ◆ Systems process input signals to produce output signals.

Slides partly extracted from “Signals and Systems”, Lecture Notes by Prof. Qing Hu, MIT, 2004, and Prof. Linshan Lee, NTU, 2009

Typical examples of signals/systems

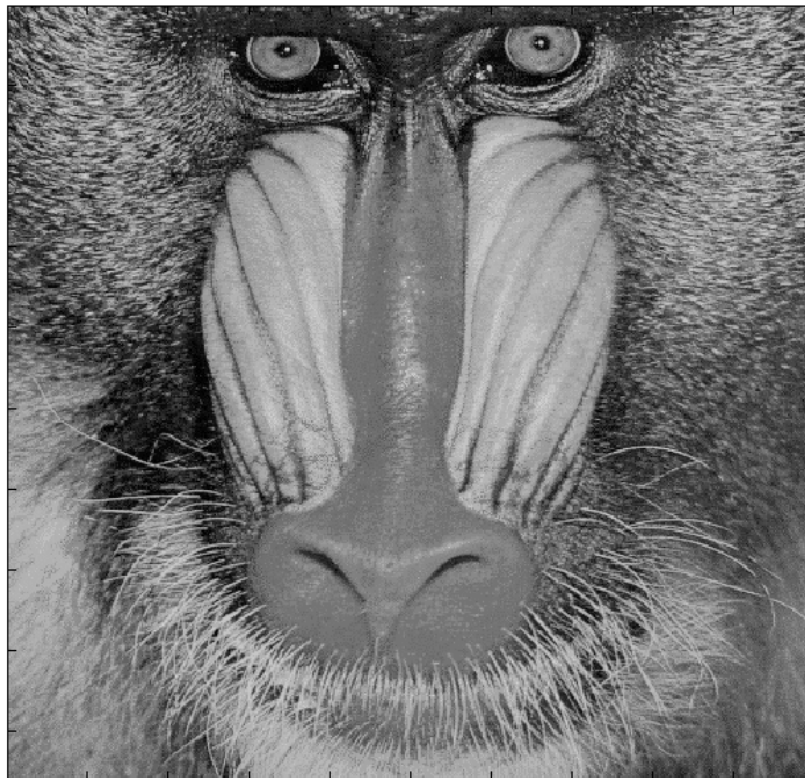
1) Communication systems



Typical examples of signals/systems

2) Image

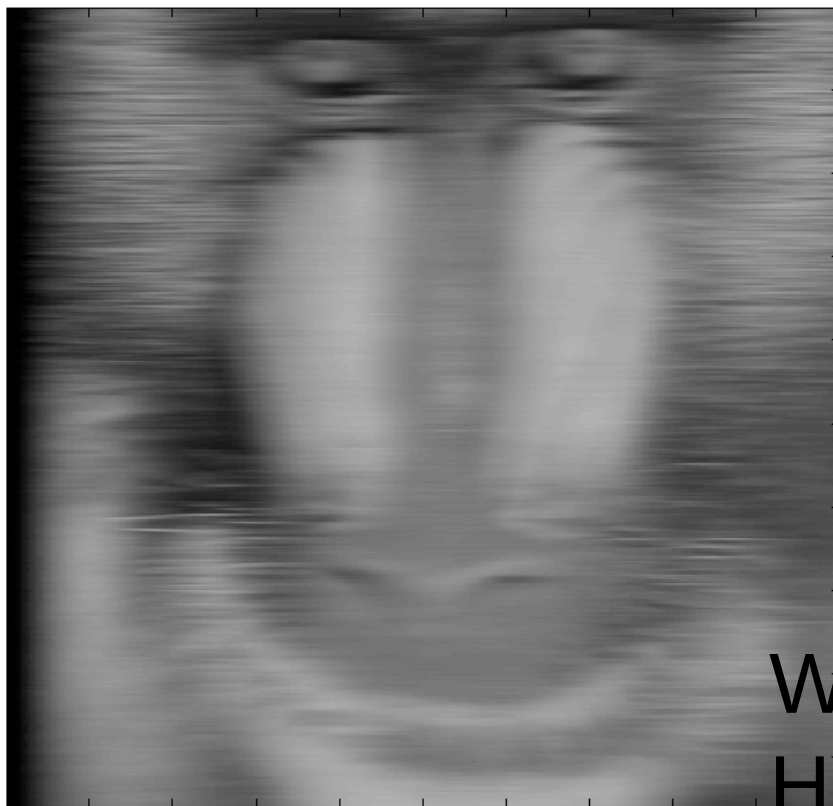
- Unblurred Image & No Noise



Typical examples of signals/systems

2) Image

- Blurred Image (bad focus)

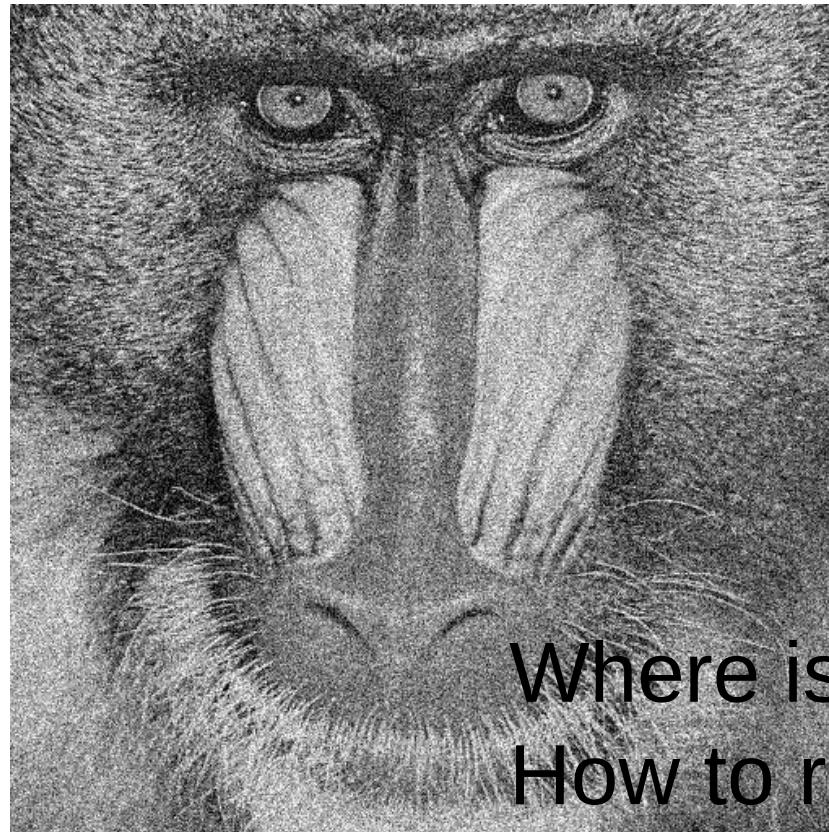


Why blurred?
How to recover?

Typical examples of signals/systems

2) Image

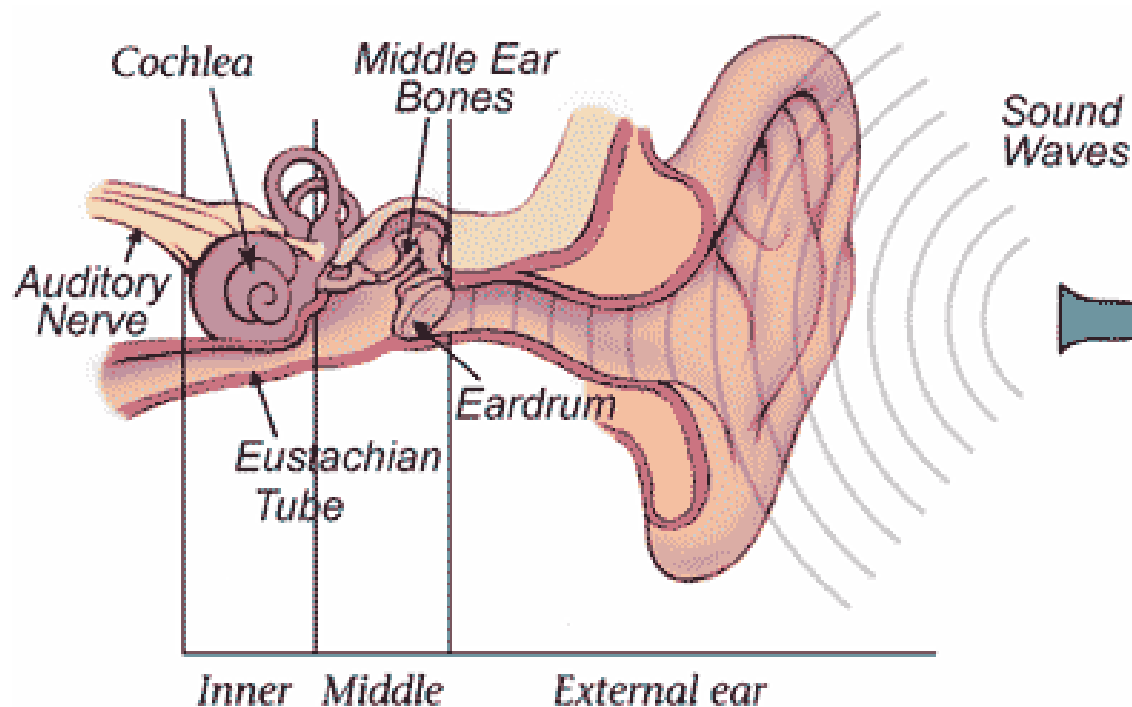
- Unblurred Image – With Noise



Where is noise added?
How to remove noise?

Typical examples of signals/systems

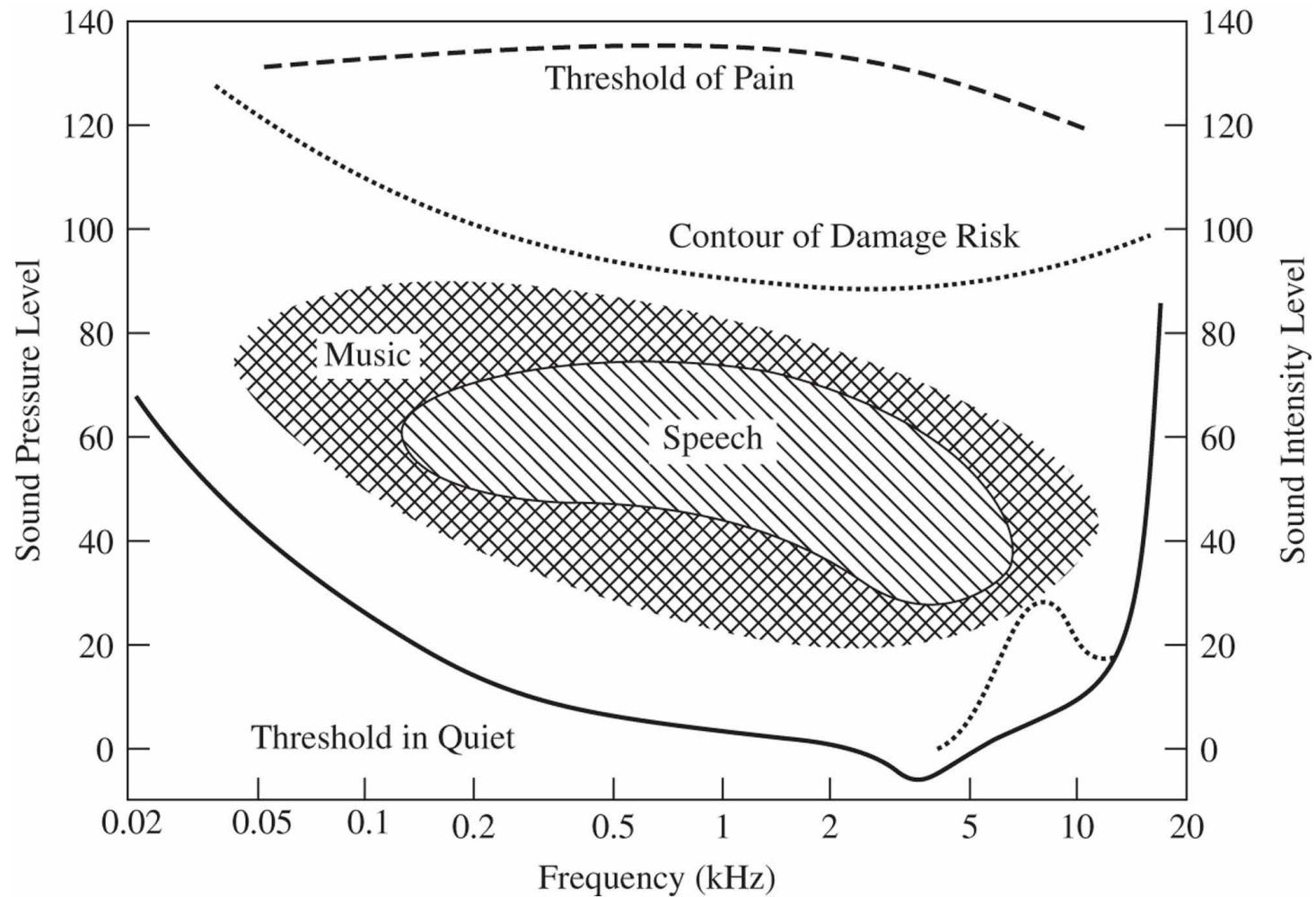
3) Human organ systems



Anatomy of the Ear

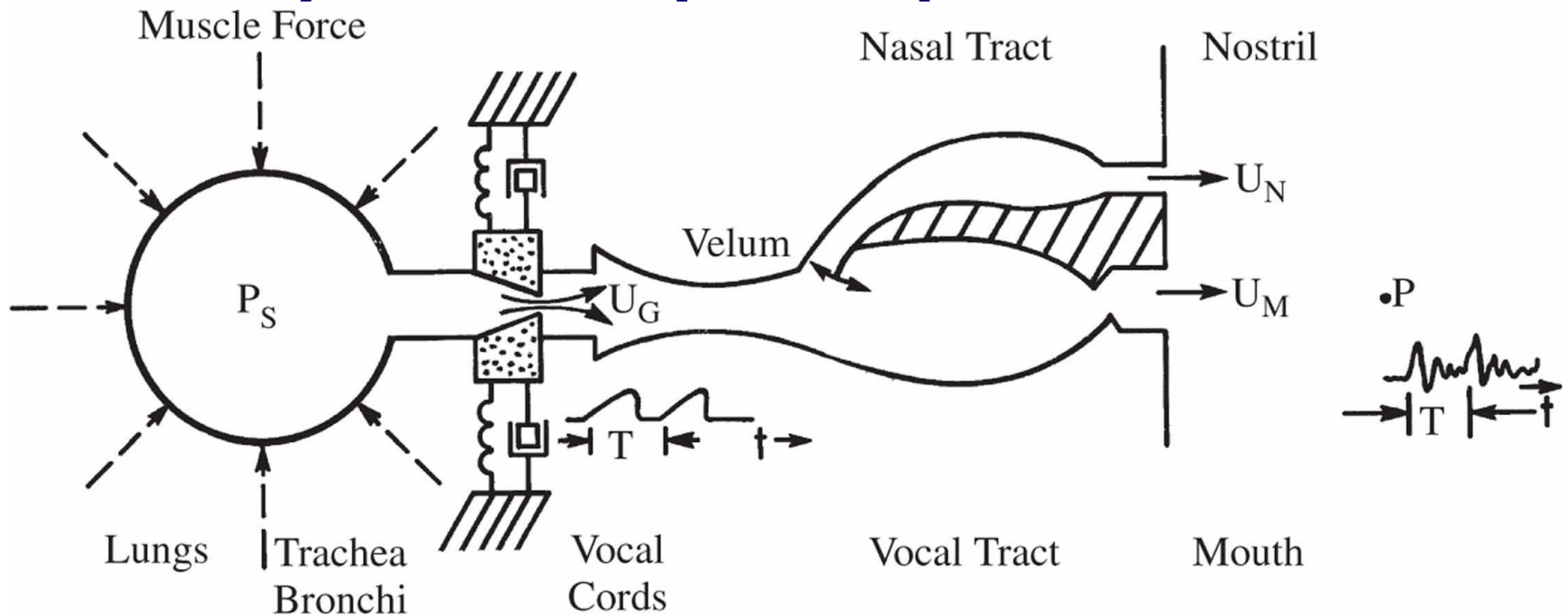
med.stanford.edu

The Range of Human Hearing

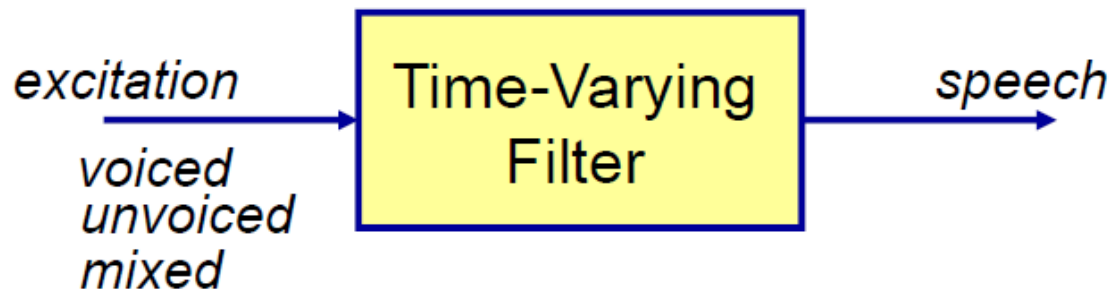


Typical examples of signals/systems

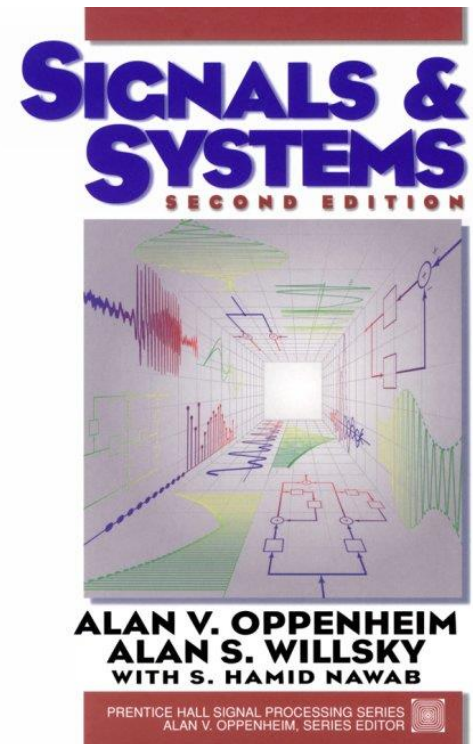
4) Human speech production



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- ◆ “Signals and Systems”, Oppenheim, Willsky and Nawab, 2nd Edition, 1997, Prentice-Hall.
- ◆ This course only teach Chapters 1 to 8.
 - Two weeks for one chapter
 - Middle-term exam for Chapters 1 to 4
 - Quiz at weeks 4 and 12
 - 平时成绩: 10%
 - 期中考试: 30%
 - 期末考试: 30%
 - 实验: 30%



Three Pillars

**Lectures
(Tutorial)**

Matlab Labs

YOU

***Assignment/Quiz
Mid-term Exam
Final Exam***

***Lab Reports
Project Report &
Presentation***

Class Schedules (cont.)

1. **Lab Session (start from week 1)**
2. **Tutorials (attend one, start from week 3)**
 - ◆ XX, 21:00-22:00
 - ◆ Room XXX, Teaching Building XX
3. **Tutorial: Every week (no for week 1)**
4. **Assignment: Every week (no for week 1)**
 - ◆ Submit assignment in the Blackboard system on Friday of the next week.
 - ◆ Late submission will have 20% reduction each day for the assignment score.

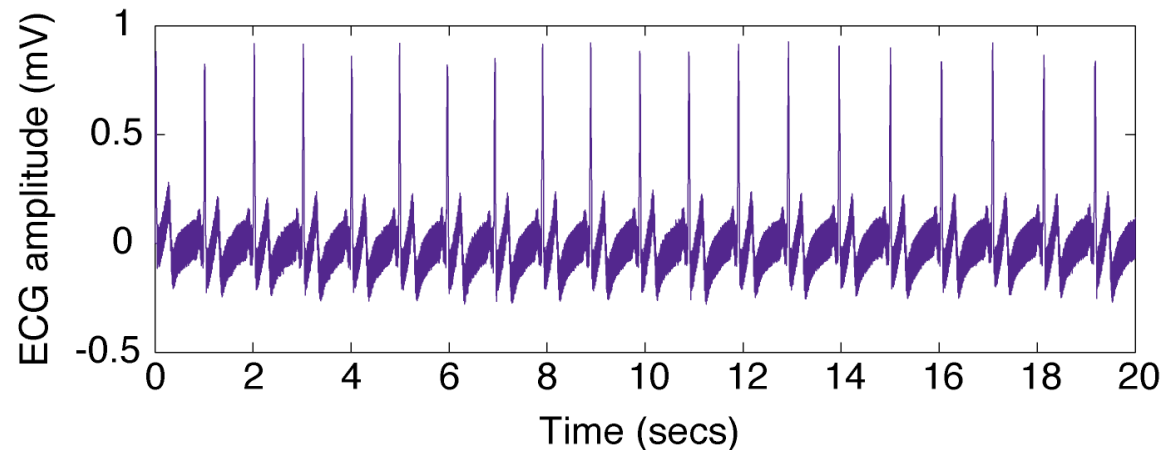
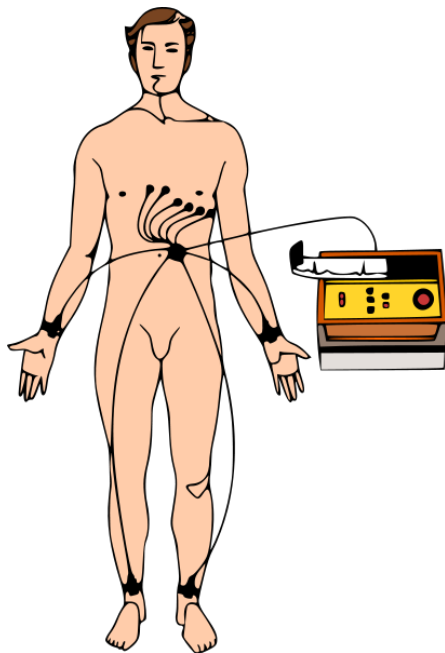
Examples of Signals: Physical Meaning

- Electrical signals – voltages and currents in a circuit
- Acoustic signals – audio or speech signals, vibration intensity of sound source
- Video signals – intensity variations in an image (e.g. a CT scan)
- Biological signals – sequence of bases in a gene (e.g., ‘...ATGGCTGA...’)
- We will treat noise as unwanted signals.

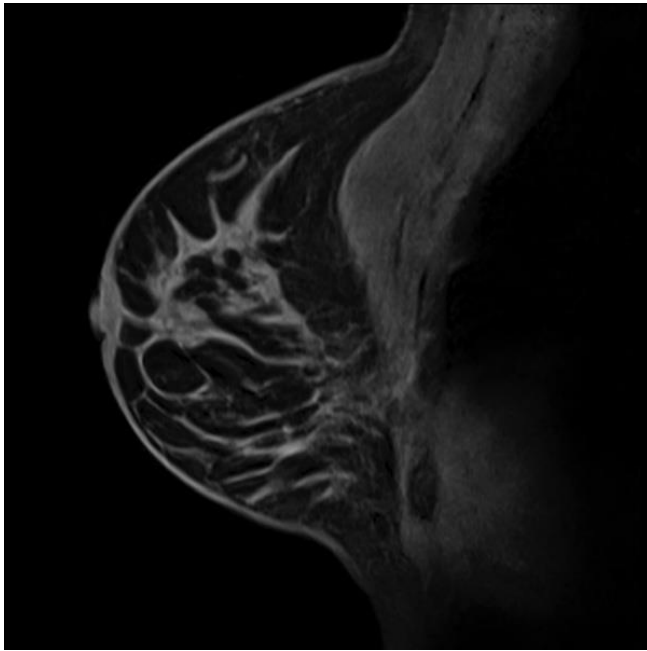
Signal Classification:

1) Type of Independent Variable

- **Time** is often the independent variable.
Example: the electrical activity of the heart recorded with chest electrodes — the electrocardiogram (ECG).



The variables can also be **spatial**

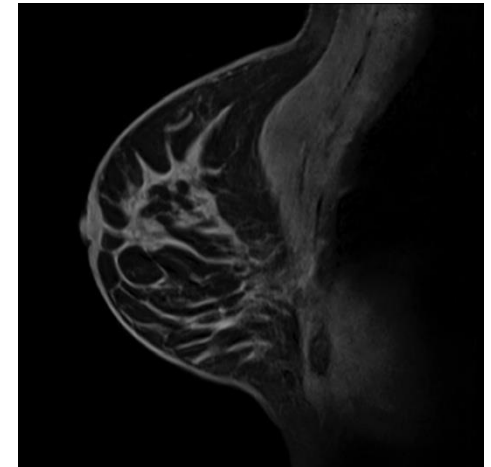
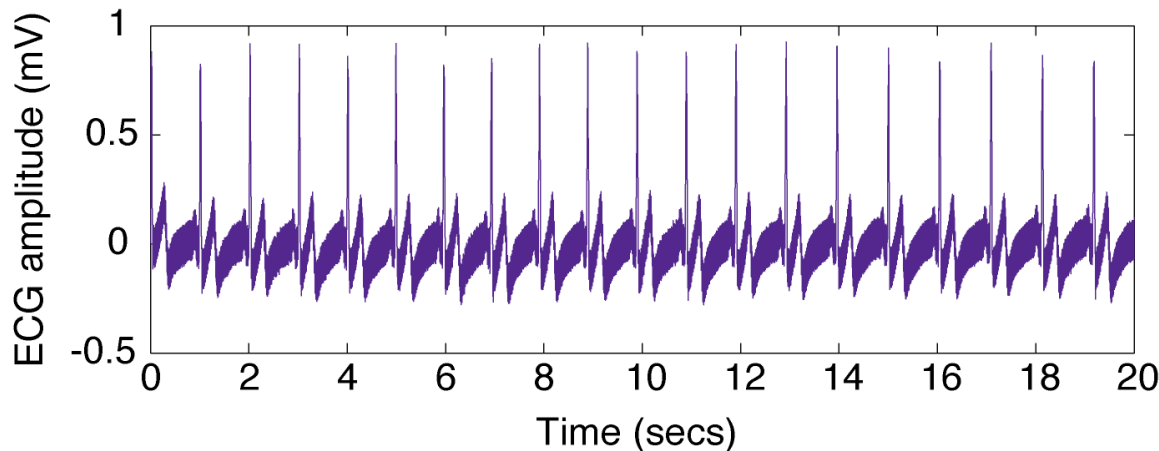


- e.g. Breast MRI

In this example, the signal is the intensity as a function of the spatial variables x and y .

Independent Variable Dimensionality

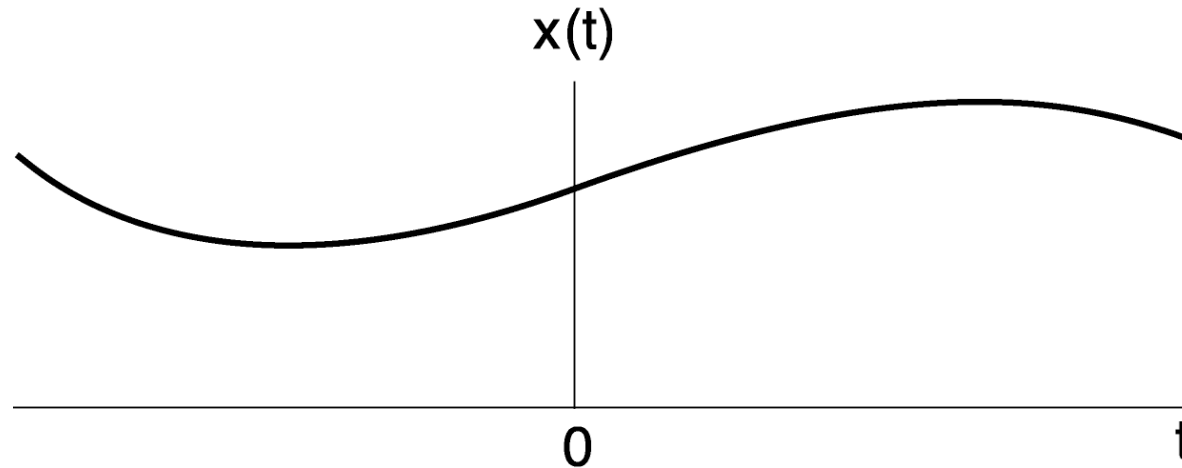
- An independent variable can be 1-D (t in the ECG), 2-D (x, y in an image), or 3-D (x, y, t in an video).



- We focus on 1-D for mathematical simplicity but the results can be extended to 2-D or even higher dimensions.
- Also, we will use a generic time t for the independent variable, whether it is time or space.

Signal Classification:

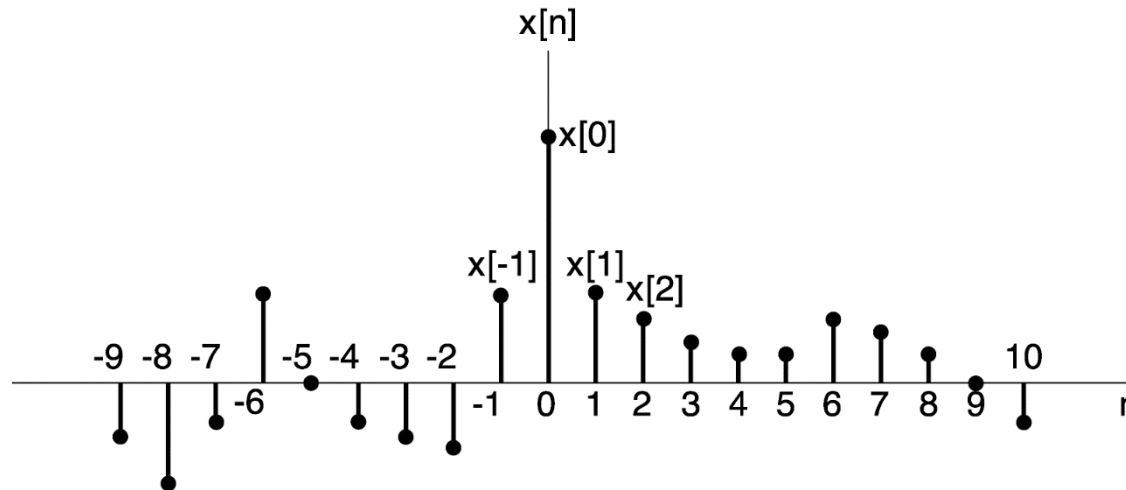
2) Continuous-time (CT) Signals



- Most of the signals in the physical world are CT signals, since the time scale is infinitesimally fine, so are the spatial scales. E.g. voltage & current, pressure, temperature, velocity, etc.

Discrete-time (DT) Signals

- $x[n]$, n — integer, time varies discretely



- Examples of DT signals in nature:
 - ◆ DNA base sequence
 - ◆ Population of the n th generation of certain species
 - ◆ ...
- Notation: $x(t)$ — CT, $x[n]$ — DT

Many Human-made Signals are DT



*Weekly Dow-Jones
industrial average*

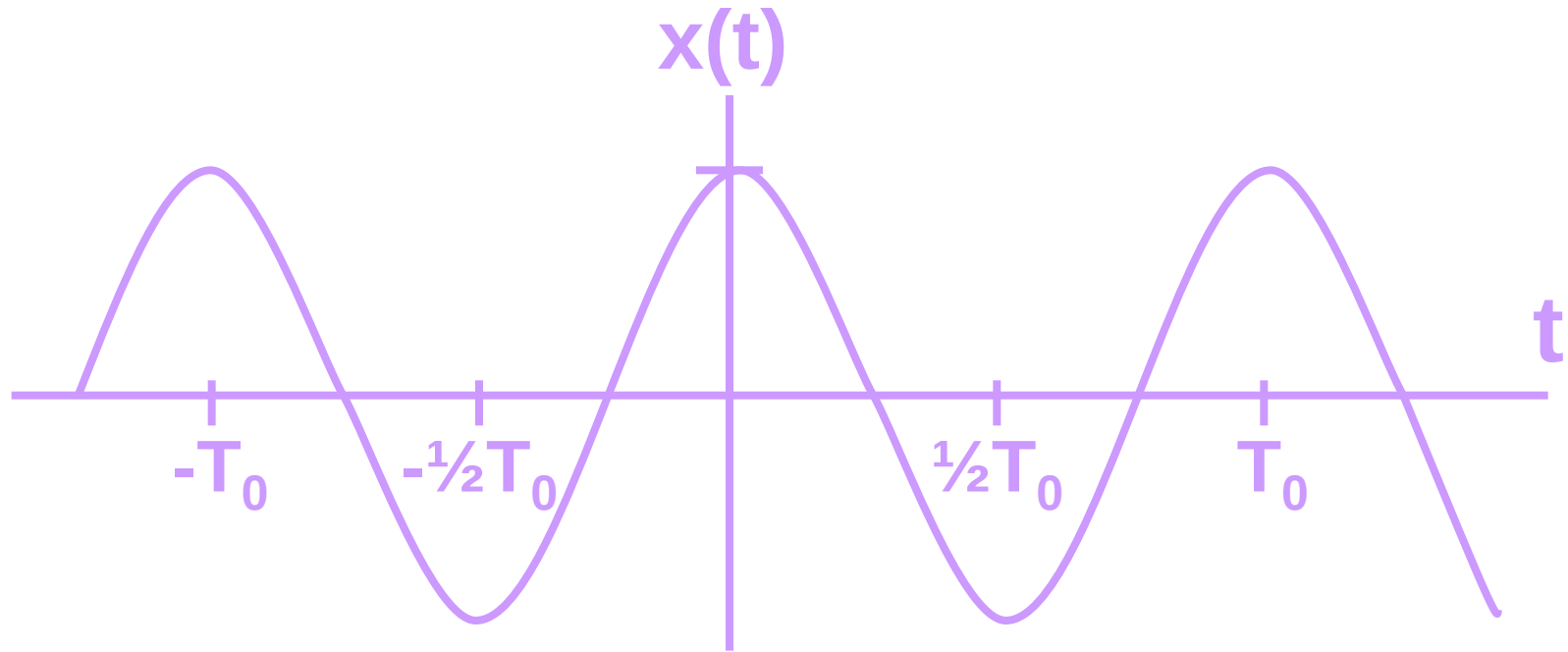


Digital image

- Why DT? — Can be processed by modern digital computers and digital signal processors (DSPs).

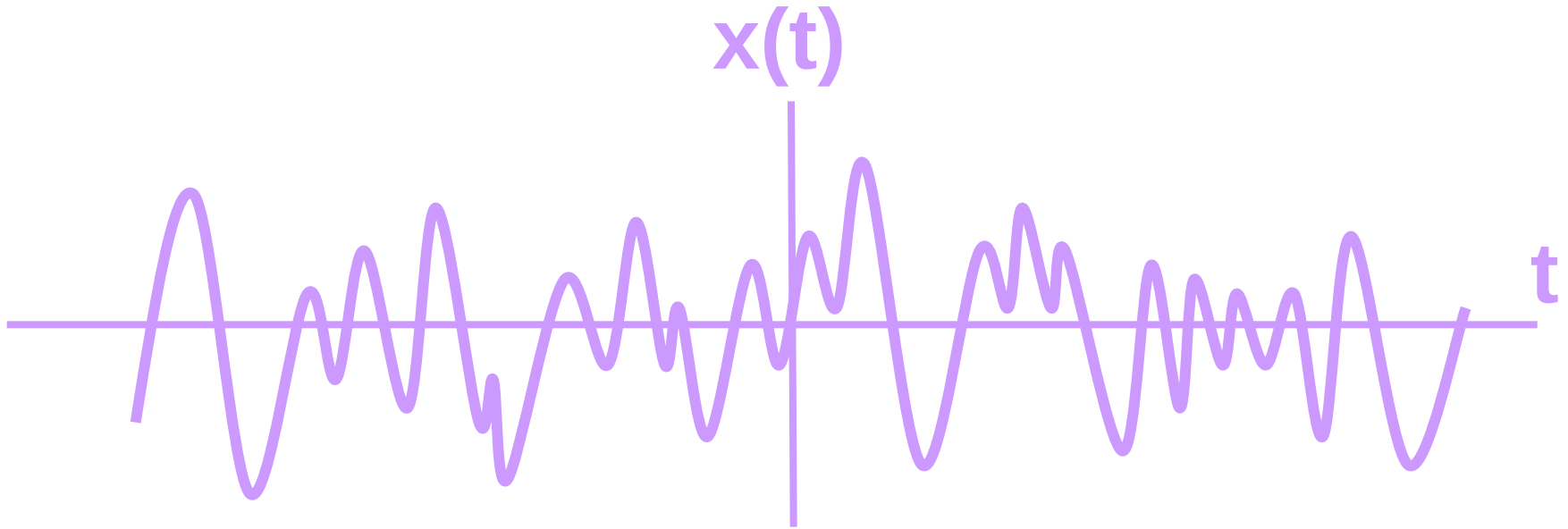
Signal Classification:

3) Deterministic Signal



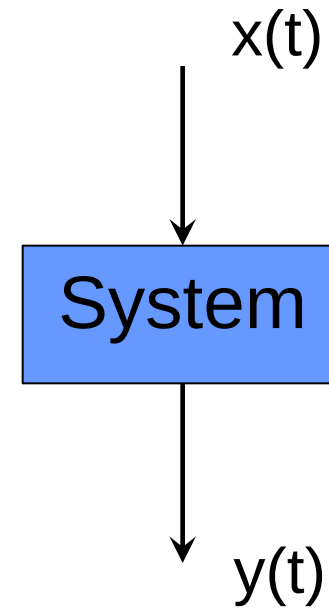
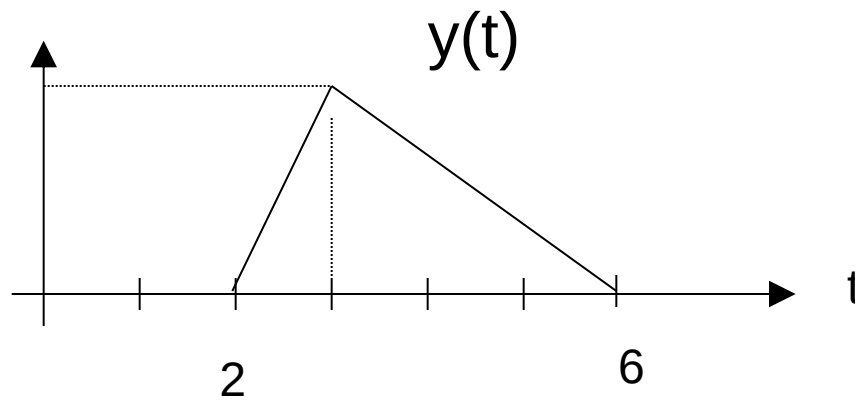
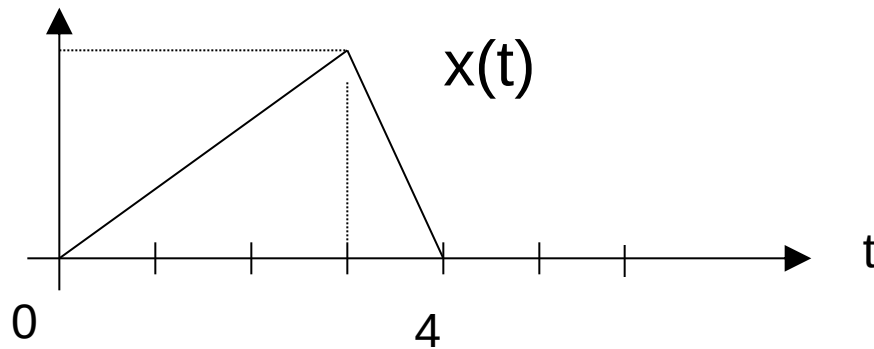
- Each value of the signal is fixed, and can be determined by a mathematical expression, rule, or table.
- Future values of the signal can be calculated from past values with complete confidence.

Random Signal



- Having a lot of uncertainty about its behaviour.
- Future values cannot be accurately predicted, and can usually only be guessed based on the averages of sets of signals.

Transformation of a Signal



Transformation of a Signal

- Time Shift

$$x(t) \rightarrow x(t - t_0) \quad , \quad x[n] \rightarrow x[n - n_0]$$

- Time Reversal

$$x(t) \rightarrow x(-t) \quad , \quad x[n] \rightarrow x[-n]$$

- Time Scaling

$$x(t) \rightarrow x(at) \quad , \quad x[n] \rightarrow ?$$

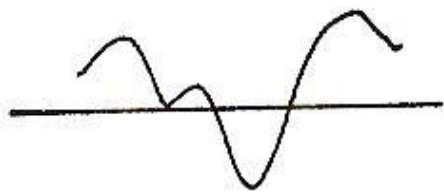
- Combination

$$x(t) \rightarrow x(at + b) \quad , \quad x[n] \rightarrow ?$$

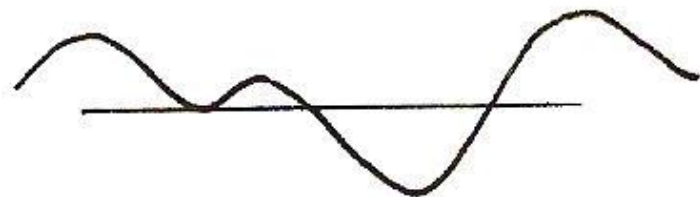
Transformation of a Signal

Time Scaling

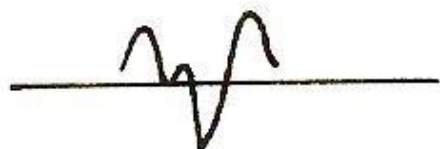
$x(t)$



$x(at), a < 1$



$x(at), a > 1$



Transformation of a Signal

Combination

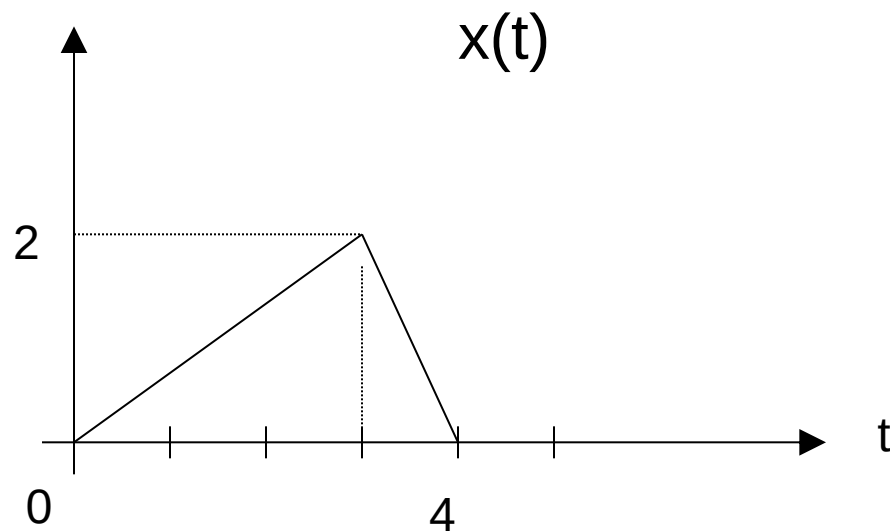
$$x(t) \rightarrow x(at + b)$$

- 1) Linearly stretched if $|a| < 1$, and linearly compressed if $|a| > 1$
- 2) Reversed in time if $a < 0$
- 3) Shifted in time if $b \neq 0$

Suggested steps:

- First delay or advance $x(t)$ with b , i.e. (3).
- Then scaling/reversing the resulting signal with factor a , i.e. (1) and (2).

Class problem



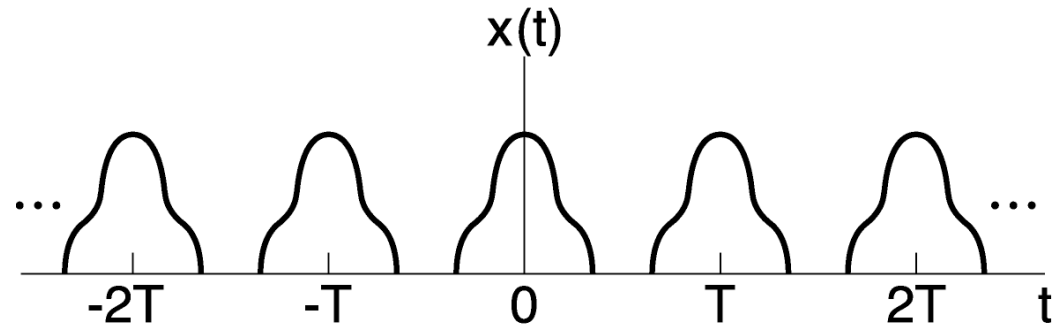
$x(-2t+2)$?

Signals with Symmetry

● Periodic Signals

◆ CT $x(t) = x(t + T)$, T : period
 $x(t) = x(t + mT)$, m : integer

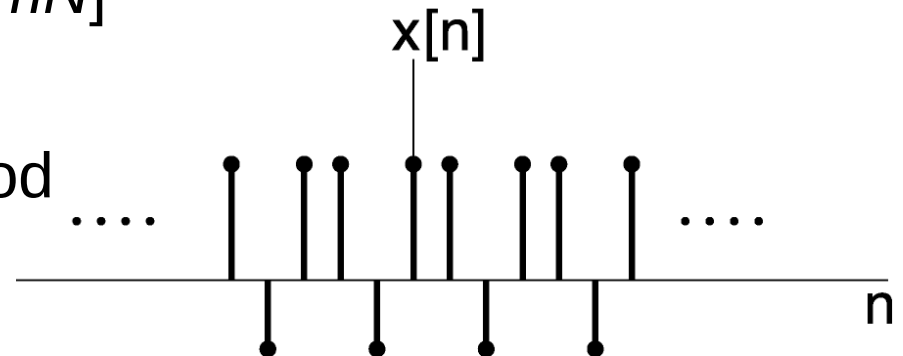
T_0 : Fundamental period, the smallest positive value of T



◆ DT $x[n] = x[n + N] = x[n + mN]$

N : period

N_0 : fundamental period



● Aperiodic: NO period

Signals with Symmetry (cont.)

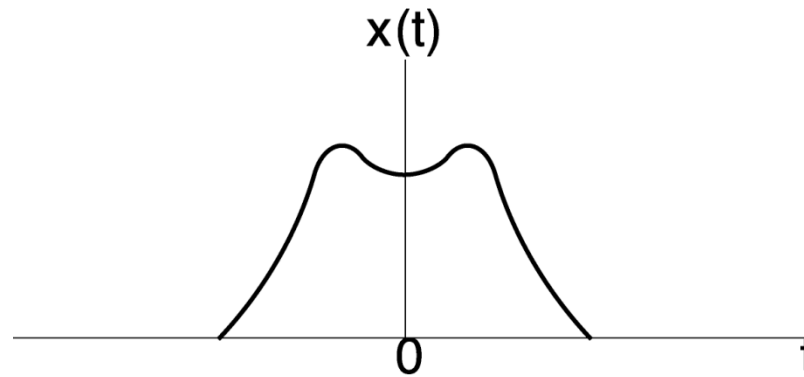
True or false

- If a signal is periodic, its duration is infinity from $(-\infty, +\infty)$.
- The period of continuous signal must be integer.

Signals with Symmetry (cont.)

- Even and Odd Signals

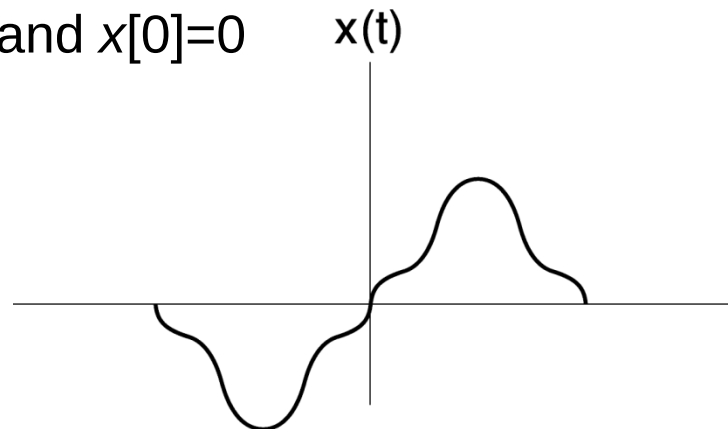
- ◆ Even $x(t) = x(-t)$ or $x[n] = x[-n]$



Example: $\cos(t)$

- ◆ Odd $x(t) = -x(-t)$ or $x[n] = -x[-n]$

- $x(0)=0$, and $x[0]=0$



Example: $\sin(t)$

Signals with Symmetry (cont.)

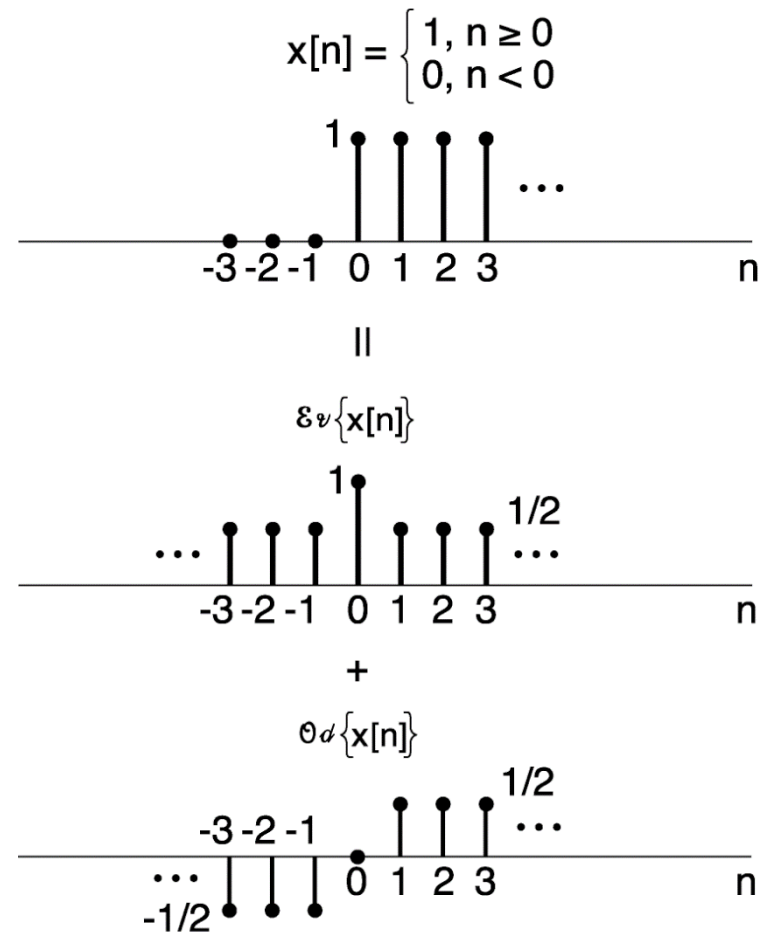
- Any signals can be expressed as a sum of *Even* and *Odd* signals. That is:

$$x(t) = x_{\text{even}}(t) + x_{\text{odd}}(t),$$

where:

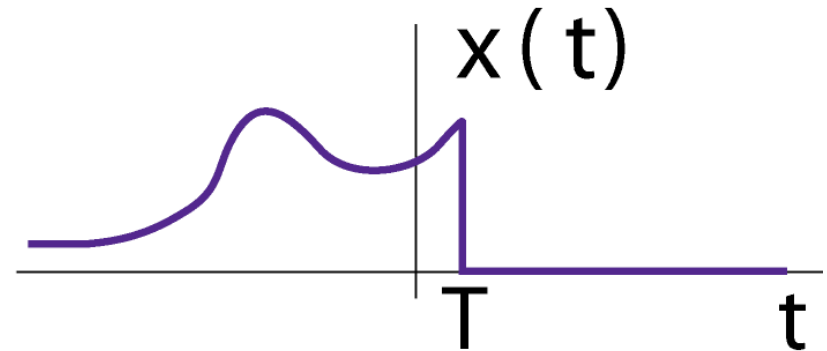
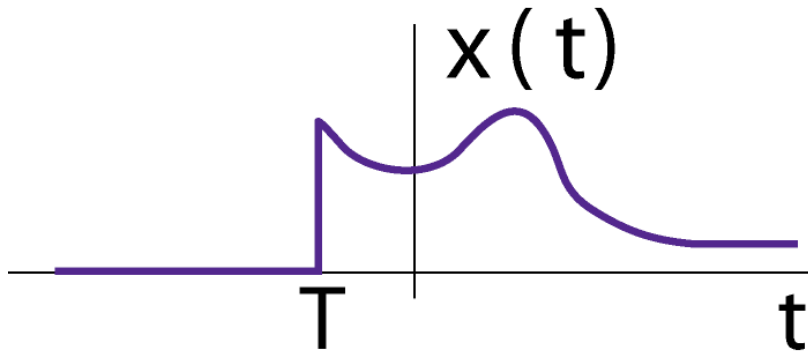
$$x_{\text{even}}(t) = [x(t) + x(-t)]/2,$$

$$x_{\text{odd}}(t) = [x(t) - x(-t)]/2.$$

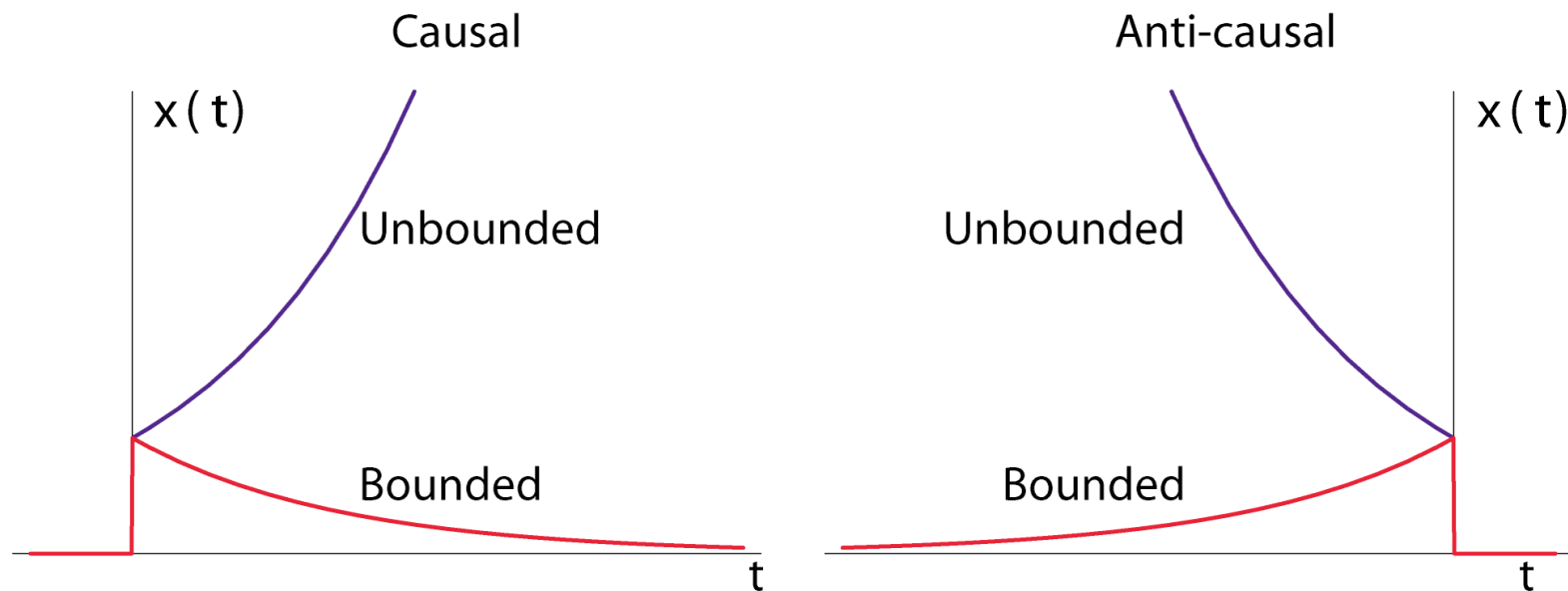


Right- and Left-Sided Signals

- A right-sided signal is zero for $t < T$, and
- A left-sided signal is zero for $t > T$, where T can be positive or negative.



Bounded and Unbounded Signals



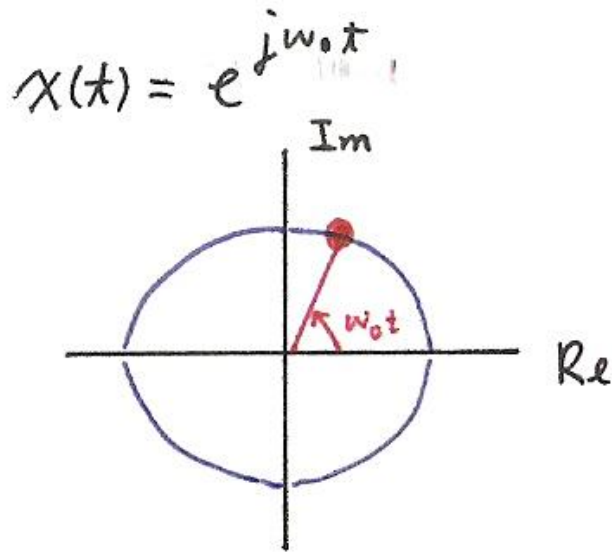
- Whether the output signal of a *system* is bounded or unbounded determines the **stability** of the system.

Exponential Signals

- A very important class of signals is presented as:
 - ◆ CT signals of the form $x(t) = e^{st}$
 - ◆ DT signals of the form $x[n] = z^n = e^{\beta n}$, $z = e^{\beta}$
where s and z are **complex** numbers.
- For both *exponential* CT and DT signals, x is a complex quantity and has:
 - ◆ a **real and imaginary** part [i.e., *Cartesian form*], or equivalently
 - ◆ a **magnitude and a phase** angle [i.e., *polar form*].
- We will use whichever form that is convenient.

Periodic Complex Exponential/Sinusoidal Signals

when $s=j\omega_0$

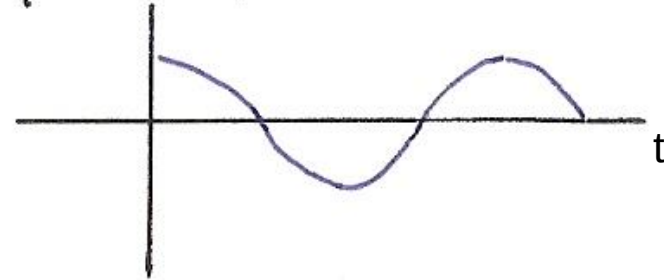


Euler's relation

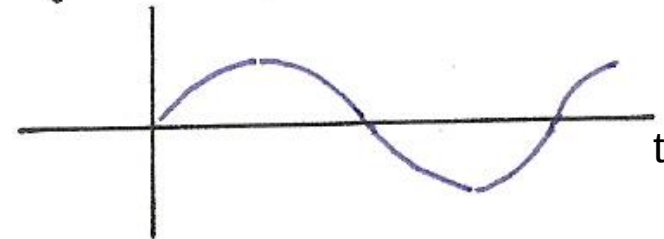
$$e^{jx} = \cos x + j \sin x$$

$\omega_0 t$ is defined as phase

$$\operatorname{Re} \{ e^{j\omega_0 t} \} = \cos \omega_0 t$$



$$\operatorname{Im} \{ e^{j\omega_0 t} \} = \sin \omega_0 t$$

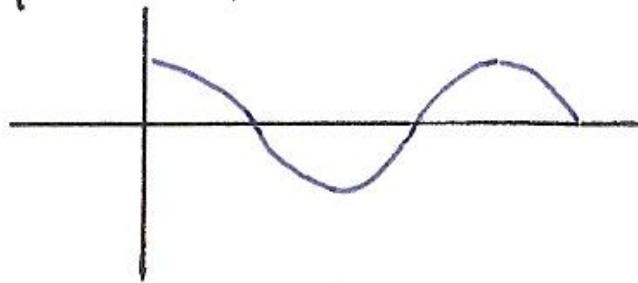


Real and imaginary parts are periodic signals with the same period, but **out of phase** (90° phase difference)

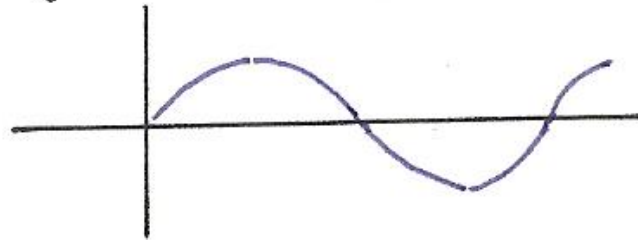
Periodic Complex Exponential/Sinusoidal Signals

$$x(t) = e^{j\omega_0 t} = \cos(\omega_0 t) + j \sin(\omega_0 t)$$

$$\operatorname{Re} \{ e^{j\omega_0 t} \} = \cos \omega_0 t$$



$$\operatorname{Im} \{ e^{j\omega_0 t} \} = \sin \omega_0 t$$



- Fundamental frequency: ω_0
- Fundamental period: $T_0 = \frac{2\pi}{\omega_0}$
- In CT, $e^{j\omega_0 t}$ **always** periodic
- Distinct signals for distinct values of ω_0 .
- Rapid variation with large ω_0

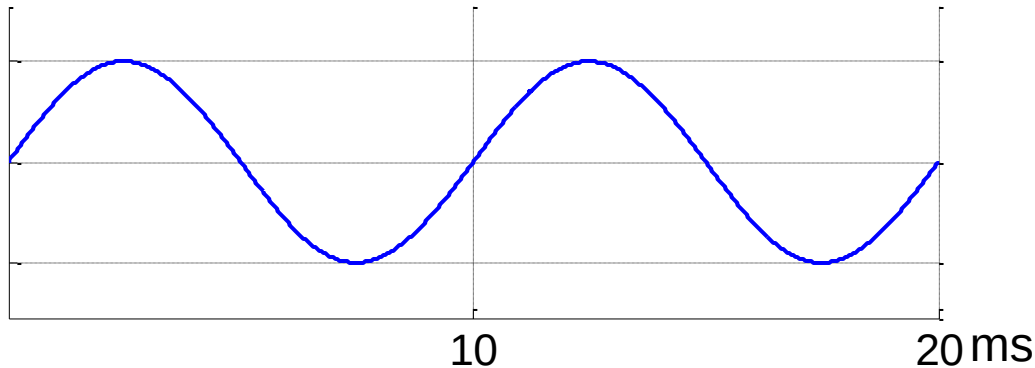
Periodic Complex Exponential/Sinusoidal Signals (cont.)

- To express sinusoidal by periodic exponentials,
e.g.,

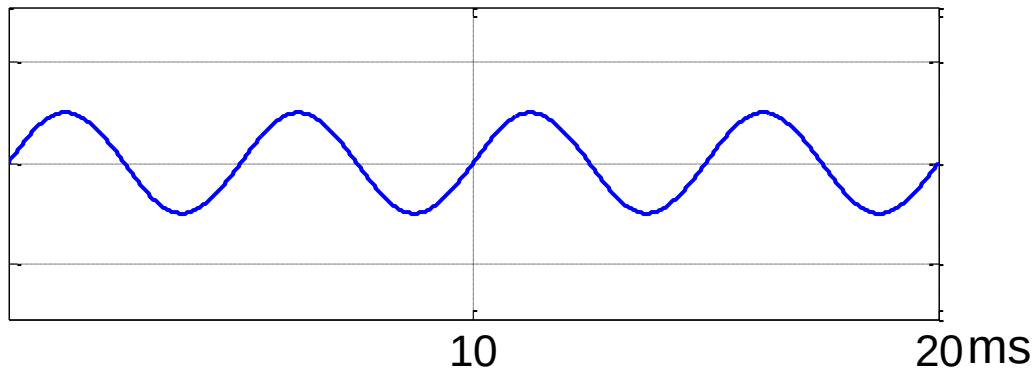
$$\cos(x) = \operatorname{Re}(e^{jx}) = (e^{jx} + e^{-jx})/2$$

$$\sin(x) = \operatorname{Im}(e^{jx}) = (e^{jx} - e^{-jx})/2j$$

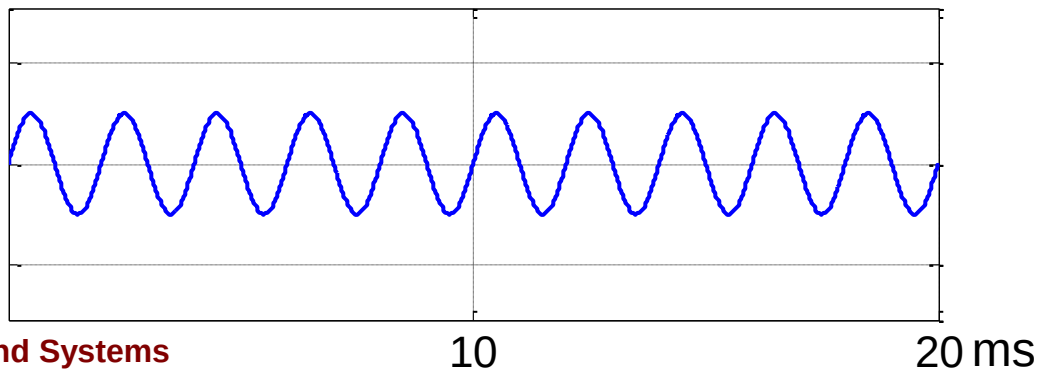
Harmonically Related Signal Sets



$$T_0 = 10 \text{ ms}$$
$$\omega_0 = 2\pi \times 100 \text{ Hz}$$



$$T_1 = 5 \text{ ms}$$
$$\omega_1 = 2\pi \times 200 \text{ Hz}$$



$$T_2 = 2 \text{ ms}$$
$$\omega_2 = 2\pi \times 500 \text{ Hz}$$

Harmonically Related Signal Sets (cont.)

- A set of periodic exponentials which have a common period T_0 .

$$\{\phi_k(t) = e^{jk\omega_0 t}, k = 0, \pm 1, \pm 2, \dots\}$$

fundamental frequency $|k\omega_0|$

must be integer multiple

fundamental period $T_k = \frac{2\pi}{|k\omega_0|} = \frac{T_0}{|k|}, \quad T_0 = \frac{2\pi}{\omega_0}$

- The k th harmonic $\phi_k(t)$ is periodic with period T_0 , as it goes through $|k|$ of its fundamental periods T_k in duration of length T_0 .

Summary of week 1

- **Meaning of signals and systems**
- **How to describe signals?**
- **Transformation of a signal**
- **Signal properties**
- **Periodic complex exponential signal**
 - ◆ **Harmonically related signal set**